Quotient rule

1 To differentiate $y = \frac{3x+4}{4x-3}$ using the quotient rule, let $u = 3x + 4$ and $v = 4x - 3$, then:

- **a** Find u'.
- **b** Find v'.
- **c** Hence, find y'.
- **d** Is it possible for the derivative of this function to be zero?

2 To differentiate $y = \frac{5x}{8x-9}$ using the quotient rule, let $u = 5x$ and $v = 8x - 9$, then:

- **a** Find u'.
- **b** Find v'.
- **c** Hence, find y'.
- **d** Is it possible for the derivative of this function to be zero?

3 To differentiate $y = \frac{9x^2}{2x+8}$ using the quotient rule, let $u = 9x^2$ and $v = 2x + 8$, then:

- **a** Find u'.
- **b** Find v'.
- **c** Hence, find y'.
- **d** Is it possible for the derivative of this function to be zero?

4 To differentiate $y = \frac{4x^2+3}{2x^2+5}$ using the quotient rule, let $u = 4x^2 + 3$ and $v = 2x^2 + 5$, then:

- **a** Find u'.
- **b** Find v'.
- **c** Hence, find y'.
- **d** Is it possible for the derivative of this function to be zero?

5 Differentiate each of the following:

- **a** $y = \frac{4x+3}{4x-3}$
- **b** $y = \frac{4x-3}{3x-4}$
- **c** $y = \frac{2x}{4x^2-3}$
- **d** $y = \frac{5x+3}{4x^2-1}$
- **e** $y = \frac{2-x}{7x+3}$
- **f** $y = \frac{x^2}{4x^3+1}$
- **g** $y = \frac{3x+2}{6+5x^2}$
- **h** $y = \frac{\sqrt{x}+2}{9\sqrt{x}+8}$

6 Find the gradient function of $y = \frac{x^2-5x-2}{\text{[illegible]}}$.

7 Consider the function $y = \dfrac{3}{x}$.

a By first rewriting it in negative index form, differentiate y.

b By using the substitutions $u = 3$ and $v = x$, differentiate y using the quotient rule.

c Find the value of x for which the gradient is undefined.

8 Consider the function $y = \dfrac{1}{8x+3}$.

a Differentiate the function using the chain rule.

b Differentiate the function using the quotient rule.

9 Consider the function $y = \dfrac{3-4x}{3x-4}$.

a Differentiate y.

b Is it possible for the derivative to be zero?

c Find the value of x that will make the derivative undefined.

10 For each of the following functions:

i Differentiate y.

ii Is it possible for the derivative to be zero?

iii Find the value of x that will make the derivative undefined.

iv State the region(s) of the domain for which the function is increasing.

v State the region(s) of the domain for which the function is decreasing.

a $y = \dfrac{x+6}{x-6}$ **b** $y = \dfrac{5x}{3x-4}$ **c** $y = \dfrac{4x}{5-2x}$ **d** $y = \dfrac{2x+3}{2x-3}$

11 Differentiate each of the following:

a $y = \dfrac{\left(4t^2+3\right)^3}{(5+2t)^5}$ **b** $y = \sqrt{\dfrac{2+7x}{2-7x}}$

Gradients and tangents

12 Find $f'(4)$ for $f(x) = \dfrac{5x}{16 + x^2}$.

13 Find $f'(1)$ for $f(x) = \dfrac{8}{2 + 2x^2}$.

14 Find the value of $f'(0)$ if $f(x) = \dfrac{x}{\sqrt{16 - x^2}}$.

15 Consider the function $f(x) = \dfrac{4x^9}{(x+2)^4}$.

- **a** Find $f'(2)$.
- **b** Is the function increasing or decreasing at $x = 2$?

16 Find the gradient of $f(x) = \dfrac{(x+7)^9}{(x+9)^4}$ at $x = -5$.

17 Find the gradient of the tangent to the curve $y = \dfrac{9x}{4x+1}$ at the point $\left(1, \dfrac{9}{5}\right)$.

18 Find the equation of the tangent to $y = \dfrac{x}{x+4}$ at the point $\left(8, \dfrac{2}{3}\right)$.

19 Find the equation of the tangent to $y = \dfrac{x^2 - 1}{x + 3}$ at the point where $x = 4$.

20 Consider the function $g(x)$ defined as $g(x) = \dfrac{f(x)}{x^3 + 3}$, where $f(x)$ is a function of x. Given that $f(2) = 5$ and $f'(2) = 6$, determine the value of $g'(2)$.

21 Find the values of x such that the gradient of the tangent to the curve $y = \dfrac{6x - 1}{3x - 1}$ is -3.

22 Differentiate $y = \dfrac{x^2}{x+3}$ and find the value of a if $y' = 0$ at $x = a$.

23 Differentiate $y = \dfrac{x^2 + k}{x^2 - k}$ and find the possible values of k given that $y' = 1$ at $x = -3$.